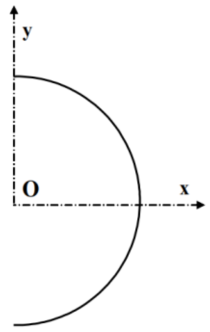
Geometry of Mass

Exercise 01: Demi-circumference

Let there be a semicircle of radius R, centered at O and with linear mass.

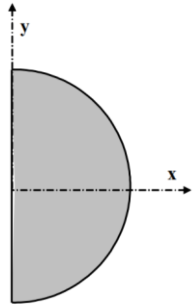
Determine the position of the center of gravity G.



Exercise 02: Half disk

Let there be a semicircle of radius R, with center O and surface mass.

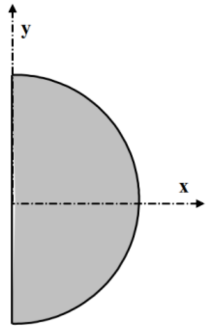
Determine the position of the center of gravity G.



Exercise 03: Hemisphere

Let there be a hemisphere of radius R, centered at O and with a density.

Determine the position of the center of gravity G.



Exercise 04: Weight barrage

A concrete gravity dam, with a triangular cross-section, rests on the ground and creates a water reservoir of height h.

NB: It is located in the middle of the dam in the width direction (followingy).

1- Give the surface S of a section of the dam. Retrieve this result by integrating a small element of surface: $S = \int_s \; ds$

2- Determine the position of the center of gravity G.